

Sensation

CT

Troubleshooting Guide

System

Troubleshooting Guide WCS (water/air/split)

This document is valid for:
Sensation 10/16/Cardiac/40/64/Cardiac64/Open

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1	General	4
	Safety information.	4
	Tools	5
2	WCS water-air-split model	6
	Overview.	6
	Primary water circuit.	7
	Secondary water circuit	8
	Refrigeration circuit.	9
	Electronic-box (indoor unit).	10
	WCS Controller board N1	10
	Circuit breakers and contactors	13
	Connectors	14
	System control table T00	15
	Display the system control table T00	15
	Normal condition of WCS-relevant parameters in T00	17
	Error messages	19
	Troubleshooting the WCS.	20
	WCS power distribution	20
	Gantry temperature.	21
	Water temperature	22
	Water pressure	24
	Water flow	24
	Water pump.	25
	Cooling fan (outdoor unit)	26
	Expansion reservoir	28
	Adjustment drive of mixer unit (right gantry stand)	29
	Compressor (refrigeration circuit)	30
	High pressure condition (refrigeration circuit)	30
	Low pressure condition (refrigeration circuit)	31
	WCS controller board N1	31
3	Changes to previous version	33

Safety information

Read the following instructions as well as the most recent version of

- **General Safety Notes** (TD00-000.860.01.xx.xx)
- **Product-specific safety Information** (CT02-023.860.01.xx.xx)

before trouble shooting the WCS.

The following instructions and the above mentioned **Safety documentation** contain important safety information as well as information about tools to be used.

NOTE

The refrigeration/chiller circuit (indoor unit) contains a cool-ant media.

Only qualified service personnel (e.g. aircondition service) is allowed to perform service and repair work in the refrigeration circuit.

=> Contact a qualified local aircondition service for service work.

=> A recommended service contact list can be found in the spare parts catalogue under the WCS component.

NOTE

The primary water circuit between the indoor unit (WCS cabinet) and the outdoor unit (fan cooler) contains antifreeze (40% glycol).

Therefore the replacement of parts in the primary water circuit that require a draining and refilling of the water circuit has to be executed by qualified service personnel (e.g. air-condition service).

Antifreeze-enhanced cooling water must be disposed accordingly.

NOTE

Operating the WCS water pump without water (e.g. during filling/refilling the gantry) will destroy the pump.

=> Do not operate the water pump without water!

NOTE

Only use the <READ> button in the function <Table load/modify> (e.g. for checking the system control table T00) !

All other functions are not necessary for service. They are normally for development staff only!

=> Improper handling can result in system malfunctions.

Tools

- Standard tool kit
- 2 water hoses with 1/2" connections for filling and draining the system
- Manometer (included with the delivery of WCS)
- Absorbent cloth

Overview

The Water Cooling System (WCS) provides the cooled water for gantry cooling. The air temperature inside the gantry is regulated within the required range (22C...30C).

Any malfunction of the WCS will normally cause error messages initiated by the MCU (master control unit). As a consequence a WCS problem may lead to image quality problems.

This document provides additional troubleshooting information if the information in the detailed error message text and/or the functional descriptions is insufficient.

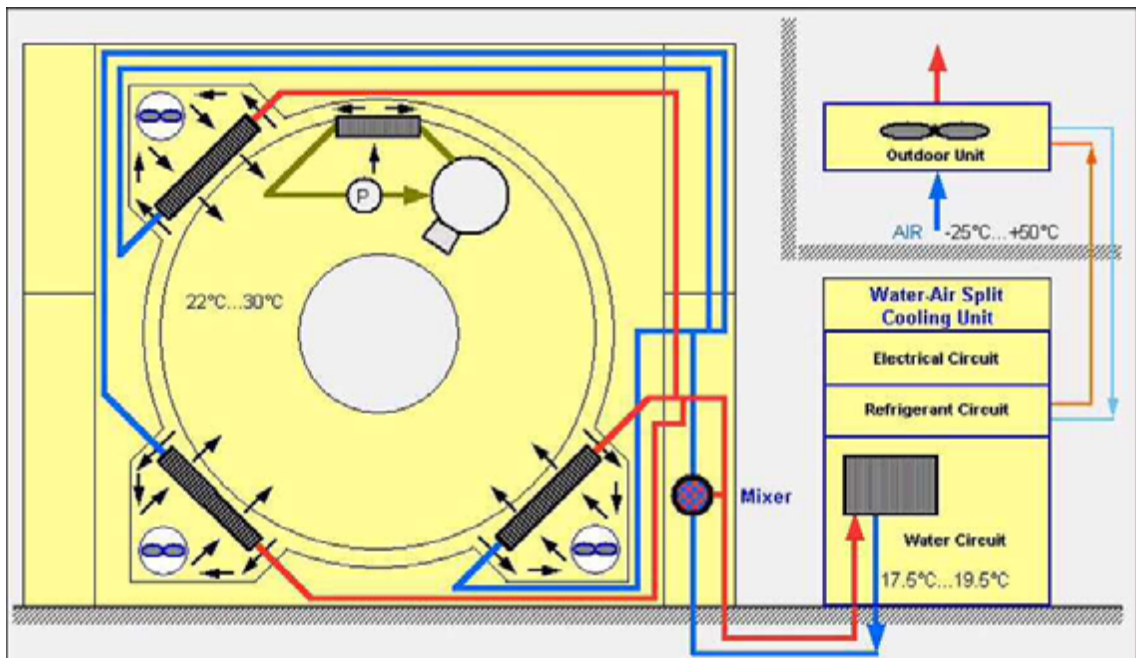


Fig. 1: Overview - WCS -w-a-Split system

The WCS water-air-split model contains of 2 main components:

- Indoor unit (WCS cabinet)
- Outdoor unit (fan cooler)

The Indoor unit serves as the main water cooling component, that contains 3 cooling circuits:

- Primary water circuit = Cooling circuit between indoor unit and outdoor unit.
- Secondary water circuit = Cooling circuit between the indoor unit and the gantry.
- Refrigeration circuit = Chiller circuit that produces cold water for gantry cooling and the electrical circuit for the indoor and outdoor unit.

Primary water circuit

The primary water pump (indoor unit) transports warm water (generated from the refrigeration process) to the outdoor unit. The outdoor unit acts as a water-air heat exchanger. It contains a large water cooler and a fan. It is located outside the building (e.g. on the flat roof or ground floor).

NOTE

The primary water circuit between the indoor unit (WCS cabinet) and the outdoor unit (fan cooler) contains antifreeze (40% glycol).

In the SPC, find the order# of the refill coolant media “Glykosol N”. The unit initially is filled with 60% water and 40% “Glykosol N”. Avoid mixing coolant media of different manufacturers. The water must be of drink water quality, otherwise, de-ionized water has to be used.

Therefore the replacement of parts in the primary water circuit that require a draining and refilling of the water circuit has to be executed by qualified service personnel (e.g. air-condition service).

Antifreeze-enhanced cooling water must be disposed accordingly.

Secondary water circuit

The secondary water circuit is a closed water cycle that pumps chilled water (generated by the refrigeration circuit) to the gantry for cooling. The mixer in the right gantry stand regulates the chilled water supply to the gantry

NOTE

No antifreeze is allowed in the cooling water of the secondary water circuit (indoor unit <=> gantry) !

Refrigeration circuit

The refrigeration circuit is a chiller that produces cold water for the secondary water circuit (gantry cooling). The heat generated by the chiller circuit (compression process) is removed through the primary water circuit and the outdoor unit.

NOTE

The refrigeration/chiller circuit (indoor unit) contains a coolant media.

In the SPC, find the order# of the refill coolant media “Glykosol N”. The unit initially is filled with 60% water and 40% “Glykosol N”. Avoid mixing coolant media of different manufacturers. The water must be of drink water quality, otherwise, de-ionized water has to be used.

Only qualified service personnel (e.g. aircondition service) is allowed to perform service and repair work in the refrigeration circuit.

=> Contact a qualified local aircondition service for service work.

=> A recommended service contact list can be found in the spare parts catalogue under the WCS component.

Electronic-box (indoor unit)

The WCS electronic-box contains all electronic components that control the indoor unit and the outdoor unit.

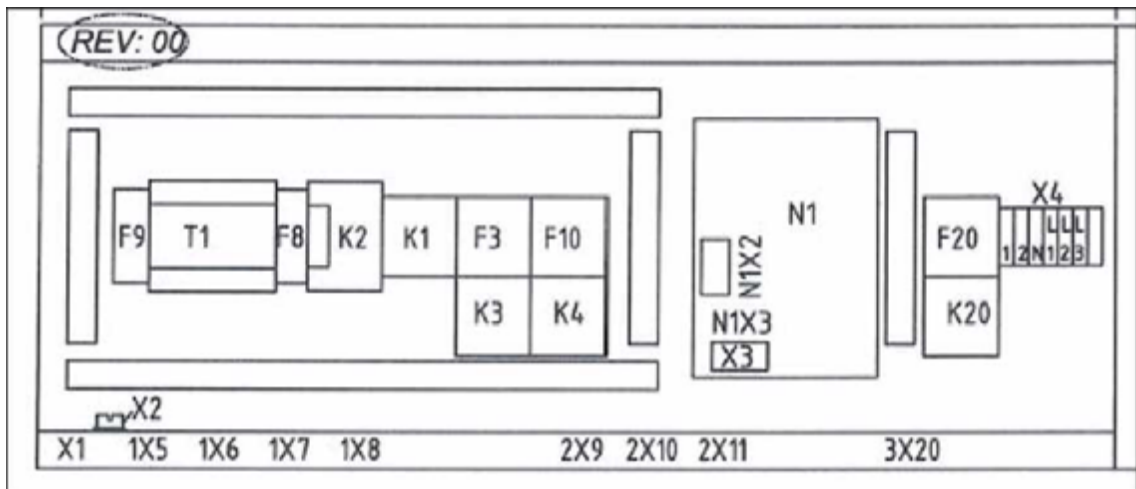


Fig. 2: Electronic box: Layout

Rev01 of the electronics box also contains capillary tube controller B1:

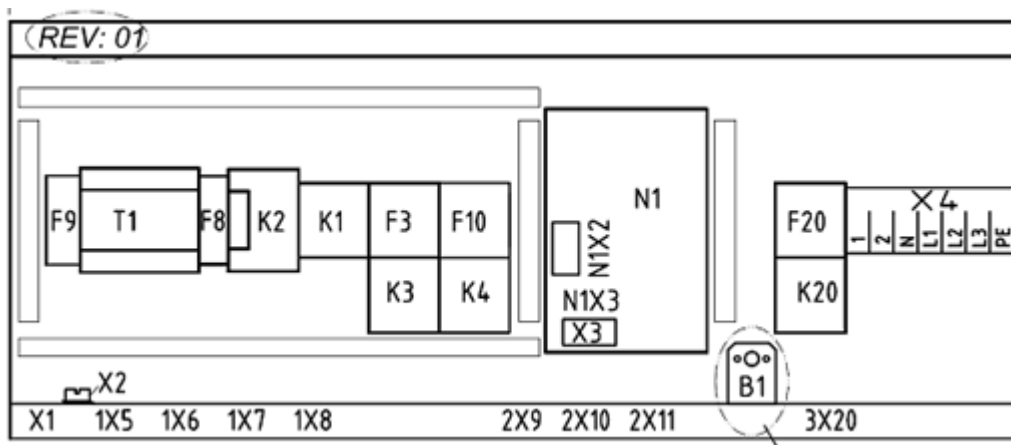


Fig. 3: Electronics box: Layout Rev01

WCS Controller board N1

The WCS controller board N1 contains several status indicators like LEDs, switches and fuses which can give a quick overview of the current condition of major WCS functions.



Fig. 4: N1 - WCS Controller (schematic layout)

Status indicators (LEDs) on N1

The WCS controller N1 in the electronic box (indoor unit) contains the following LEDs (L1...L24) that indicate the status of the WCS.

Tab. 1 LED's on the WCS controller board N1

LED-ID / color	Normal status	Function
L1 / green	blinking	Controller active (show life)
L2 / yellow	off	on, if Service switch S4 is active
L3 / n.a.	n.a.	not mounted
L4 / n.a.	n.a.	not mounted
L5 / green	on, if -->	Communication active (telegram traffic) off, if communication was interrupted
L6 / n.a	n.a.	not mounted
L7 / yellow	off	on, if --> any malfunction occurs (e.g. temperature, pressure errors) back to off, if malfunction disappeared (e.g. temp./pressure ok again)
L8 / green	off	not used
L9 / green	on, if -->	request for switching on primary water pump

LED-ID / color	Normal status	Function
L10 / green	on	off, if pressure too low (= low pressure condition in coolant circuit, F5)
L11 / green	on	off, if pressure too high (= high pressure condition in coolant circuit, F1)
L12 / green	on	50Hz / 60 Hz indicator (no v
L13 / green	off	not used
L14 / green	on	Secondary cooling pump protection F3 / OK off, if --> over current
L15 / green	on	Primary cooling pump protection F10 / OK off, if --> over current
L16 / red	off	on, if --> MCU reset occurs
L17 / green	on, if -->	Compressor 1 (small) is active
L18 / green	on, if -->	Compressor 2 (large) is active
L19 / green	on	Secondary cooling pump is running (water to/from gantry)
L20 / green	off	not used
L21 / green	on	Primary cooling pump is running (water to/from outdoor unit)
L22 / green	off	not used
L23 / green	off	not used
L24 / green	off	not used

Switches on N1

The WCS controller N1 in the electronic box (indoor unit) contains some switches (S1...S4) that enable certain functions on the WCS.

Tab. 2 Switches on the WCS controller board N1

Switch-ID	Normal status	Function
S1	not pushed	Reset of WCS controller N1 (refer to Note below)
S2	n.a.	for R&D use only
S3	n.a.	for R&D use only
S4	off (LED #L2 = off)	Service on/off (status is displayed with LED # L2)

NOTE

Do not push switch S1 during WCS operation!

This will cause an undefined status of the WCS and will stop the WCS operation.

A restart of the WCS only will be possible after a delaytime of 3 minutes.

Fuse on N1

The WCS controller N1 in the electronic box (indoor unit) contains a FUSE that protects the board from coincidental short circuits (e.g. during installation & service activities).

Tab. 3 Fuse on the WCS controller board N1

Fuse-ID	Location	Function
FUSE	(Fig. 4 / p. 11)	Protects the N1 controller board from short circuits.

If the FUSE is defective all LEDs on the WCS controller board N1 will be OFF.

Circuit breakers and contactors

The WCS electronic-box contains also several circuit breakers , contactors, power switches and a transformer. (Fig. 2 / p. 10)

Tab. 4 Circuit breakers and cocntactors at WCS electronic box

Component-ID	Function
F8	Circuit breaker, protects secondary side (18...30VAC) of transformer T1
F9	Circuit breaker, protects primary side (230VAC) of transformer T1
T1	Transformer, 230VAC / 18..30VAC, supplies power for controller N1
K1	Contactor 24V, switches on the small compressor M1
K2	Contactor 24V, switches on the large compressor M2
K3	Contactor 24V, switches on the secondary water pump M3 (water circuit: indoor unit <=> gantry)
K4	Contactor 24V, switches on the primary water pump M6 (water circuit: indoor unit <=> outdoor unit)
K20	Contactor 230V switches on the cooling fan M20 (outdoor unit) when the small compressor is activated via K1, the thermo contact (TK) of the cooling fan is inactive, and (if the capillary tube controller B1 is installed) the primary water outlet temperature exceeds 35C.
F3	Power switch, enables and protects the power for secondary water pump
F10	Power switch, enables and protects the power for primary water pump

Component-ID	Function
F20	Power switch, enables and protects the power for cooling fan (outdoor unit)
N1	(WCS Controller board N1 / p. 10), monitors and controls all WCS functions, communication with MCU
B1	The capillary tube controller switches on the fan of the outdoor unit if the primary water outlet temperature exceeds 35C.

Connectors

The WCS electronic-box contains several connectors for cable connection to internal and external components. ([Fig. 2 / p. 10](#))

Tab. 5 Connectors at WCS electronic box

Connector-ID	Function
X1	Main power for WCS (indoor unit), distributed from PDC
X2	Protective ground (PE)
X4	Main power (for outdoor unit) from onsite power distribution panel
1X5	Power for compressor 1 (M1, small)
1X6	Power for compressor 2 (M2, large)
1X7	Power for primary water pump (M6)
1X8	Power for secondary water pump (M3)
2X9	Pressure switches (low / high pressure) in refrigeration circuit
2X10	Power for drive (M5) of bypass valve in refrigeration circuit
2X11	Temperatur sensors (R1, R2, R3)
3X20	Power to the cooling fan and thermo contact of fan motor (outdoor unit)
N1X2	RS232-Interface (service)
N1X3	MCU connection

System control table T00

The system control table T00 can be displayed on the monitor in the Service mode.
It shows the status of various WCS-relevant system parameters (Tab. 6 / p. 17).

Display the system control table T00

- From user interface, select: **Options -> Local Service.**
- In the authentication page, enter the **Service Password** and Click **OK.**
- In the Home menu, click the **Control** button.
- Click **Table load/modify:**
 - ⇒ The following page appears.

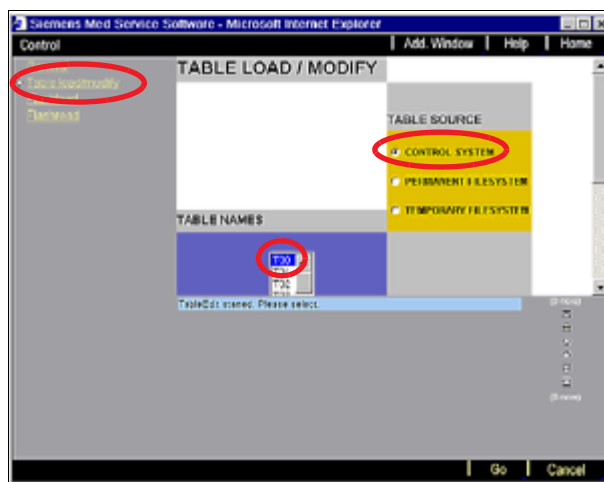


Fig. 5: Table load / modify page

- To access the control system
 - Select **T00** for TABLE NAME
 - Select **CONTROL SYSTEM** for TABLE SOURCE
 - ⇒ Click **GO**, and the following page appears

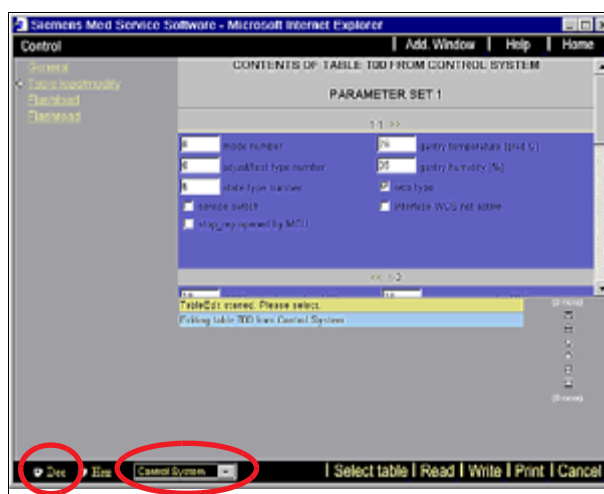


Fig. 6: Example of Parameter Set 1

- Select **Dec** (= decimal), and click **Read**
⇒ The corresponding table will be activated.

NOTE

Only use the <Read> button in the functions <Table load/modify>. All other functions are not necessary for service, and are normally for development staff only.

=> Improper handling can result in system malfunctions.

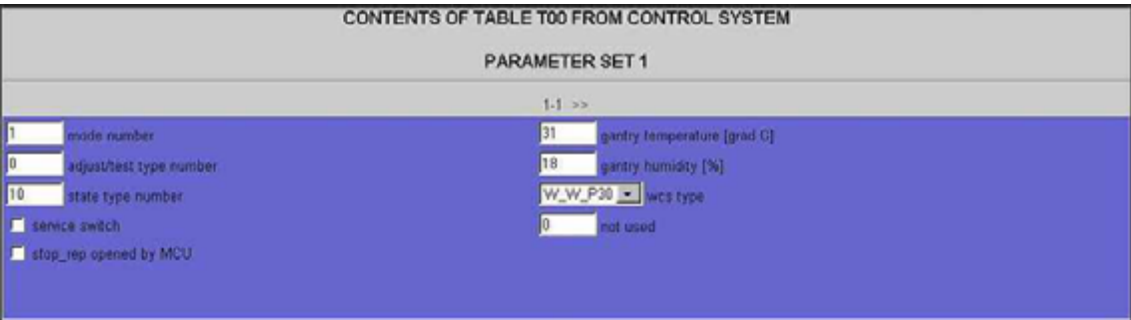


Fig. 7: System control table T00, part 1

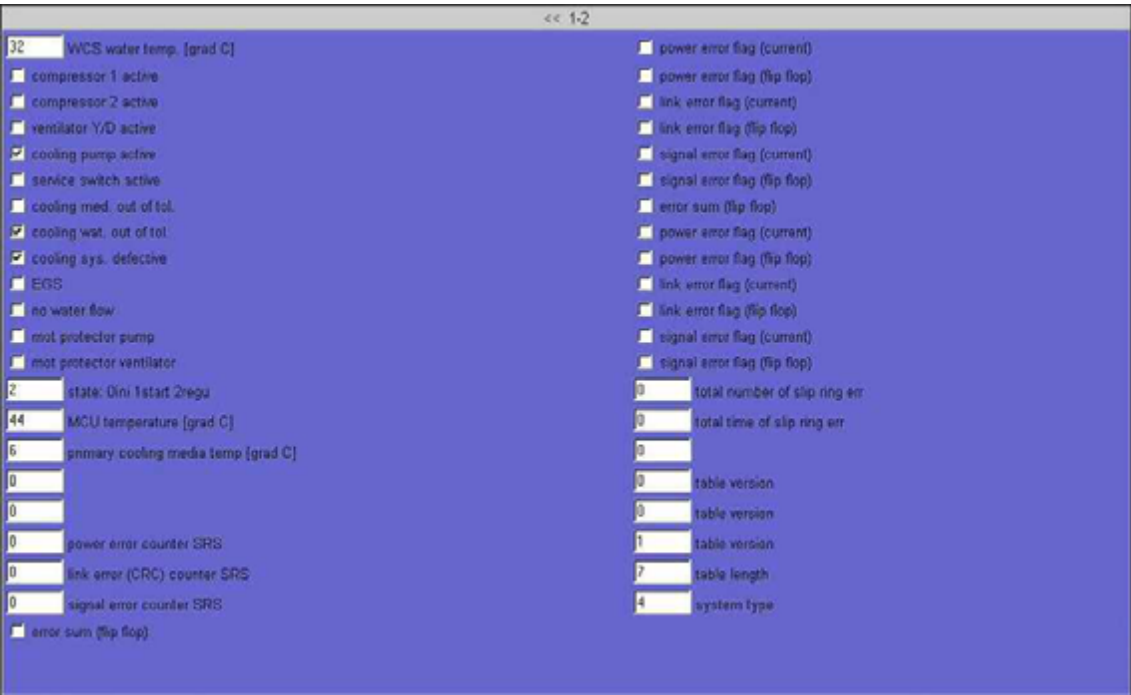


Fig. 8: System control table T00, part 2

Normal condition of WCS-relevant parameters in T00

Tab. 6 System control table T00

Parameter	Normal condition
gantry temperature (C)	= actual temperature inside gantry (22C...30C)
gantry humidity (%)	= actual humidity inside gantry
WCS water temperature (C)	= actual water temperature in secondary cooling circuit (temperature sensor R1, 18,5C +/- 1C)
compressor 1 active	On, if compressor 1 (= small type) is active
compressor 2 active	On, if compressor 2 (= large type) is active
ventilator Y/D active (WCS w-a type only !)	On
for WCS w-a-split model: = primary cooling pump active	
cooling pump active (= secondary water pump)	On
cooling media out of tolerance (= primary water circuit)	Off
cooling water out of tolerance (= secondary water circuit)	Off
cooling system defective (= refrigeration circuit)	Off, (if ON, then there was a problem in the refrigeration circuit; e.g. low/high pressure condition)
EGS (= freeze protection)	Off, (= temperature sensor R3, > 4C)
no water flow (WCS w-w model only !)	Off, (n.a. for WCS w-a-split model)
motor protector pump (= secondary pump)	Off

Parameter	Normal condition
motor protection ventilator (WCS w-a model only !)	Off
for WCS w-a-split model: = motor protection of primary pump	
primary cooling media temperature (C)	= actual water temperature in primary water circuit, water return from outdoor unit at primary pump (temperature sensor R2, 0C...50C)

Error messages

Refer to the following table of WCS-relevant error message.

Tab. 7 WCS error messages

Error code	Error message text	Severity
CT_MCU_3097	WCS internal error	Info
CT_MCU_3089	Gantry water temperature out of tolerance	Info
CT_MCU_3115	WCS primary cooling flow not OK	Warning
CT_MCU_3116	WCS primary cooling media not OK	Warning

- Check the event logbook for one or more entries of the above listed cooling error messages.
- Check the detailed error message text of the according error message.
- Follow the suggested action as mentioned in the detailed error message information.

Troubleshooting the WCS

The following procedures describe testing and troubleshooting of various cooling-relevant components and parameters.

This should enable the CSE to make a fast diagnosis of the actual status and support the detection of defects and malfunctions in WCS system.

WCS power distribution

NOTE

The power for the indoor unit (WCS cabinet) of the WCS is supplied from the PDC.

The power for the outdoor unit (fan cooler) of the WCS is NOT supplied from the PDC. It is supplied directly from the onsite power distribution panel to the indoor unit and from the indoor unit to the outdoor unit.

Indoor unit

NOTE

The power for the indoor unit (WCS cabinet) of the WCS is supplied from the PDC (power distribution cabinet).

- Open the front door of the WCS indoor unit.
- Remove the front cover from the WCS electronic box.
- Check the cable connection at X1 (electronic-box).
- Check the voltage between all 3 phases at the power switches F3 and F10 at the electronic box (indoor unit)
 - ⇒ Identical phase-to-phase voltage (400VAC/50Hz or 460VAC/60Hz) must be ON
 - ⇒ If there is no power check the circuit breaker F14 in the PDC
- Check the switch position of power switches F3 and F10
 - ⇒ F3 and F10 must be ON (= switch pos. I)
- Check the circuit breakers F8 and F9 (electronic box, indoor unit).
 - ⇒ F8 and F9 must be ON (= switch pos. I)
- Check the LEDs at the WCS controller N1 (electronic box at indoor unit)
 - ⇒ Certain LEDs must be ON (refer to LED status table: [\(Status indicators \(LEDs\) on N1 / p. 11\)](#))
- Check the FUSE on the WCS controller N1

Outdoor unit

NOTE

The power for the outdoor unit is NOT supplied by the PDC cabinet. It is supplied directly from the onsite power distribution panel to the indoor unit (X4) and from the indoor unit (3X20) to the outdoor unit.

An optional “Repair switch” may be installed outside, close to the outdoor unit, if local firecode require. If the outdoor unit doesn’t switch on, check for the existence an position of this switch.

- Check the voltage at connector X4 in the WCS electronic box (indoor unit).
 - ⇒ Identical phase-to-phase voltage (400VAC/50Hz or 460VAC/60Hz) must be ON between all 3 phases.
 - ⇒ If there is no power at X4, check the cable connection to X4 and check the power and circuit breaker at the onsite power distribution panel.
- Check the switch position at the power switch F20 in the electronic box (indoor unit)
 - ⇒ F20 must be ON (= switch pos. I)
- Check the contactor K20 in the electronics box (indoor unit)
 - ⇒ K20 must be ON if compressor M1 (small) is active, the thermo contact (TK) of the cooling fan (outdoor unit) is not active (open), and (if capillary tube controller B1 is installed) the primary water outlet temperature exceeds 35C.

Gantry temperature

NOTE

The gantry internal air temperature is monitored by two temperature sensors (B302 / B305). (1+2/Fig. 9 / p. 21).

Nominal internal gantry temperature: 26C +/-4C

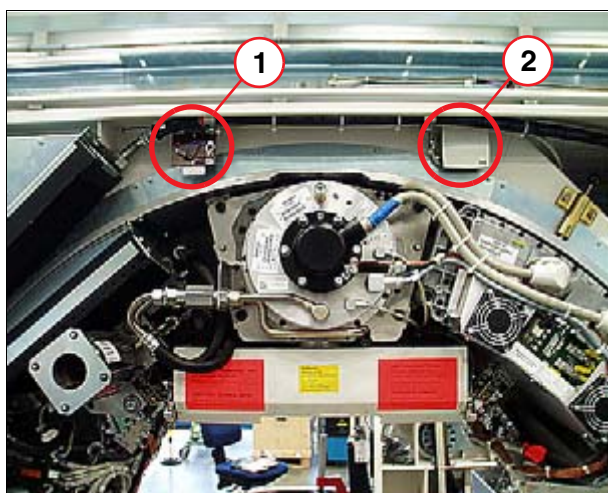


Fig. 9: Temperature Sensor in Gantry

Pos. 1 B 305

Pos. 2 B 302

Tab. 8 Temperature sensors in the indoor unit

Temp.sensor	Nominal value	Function
B302	< 35 C	Actual internal Gantry temperature: Measures the gantry internal temperature and generates an error (CT_MCU_3089) if 35C are reached. Used for regulation of the gantry water circuit (mixer)
B305	< 43 C	Gantry internal temperature limit (safety switch) PDC switches off the power to the gantry if 43C are reached.

- Check the actual gantry temperature by reading out the system control table T00 ([System control table T00 / p. 15](#))
 - ⇒ Nominal internal gantry temperature: 26C (+/- 4C)
- Check the 3 gantry cooling fans (left/right bottom corner, upper left corner).
 - ⇒ All 3 gantry cooling fans must be operating.
- Check the mixer valve in the right gantry stand ([Adjustment drive of mixer unit \(right gantry stand\) / p. 29](#))
- Check the secondary water pump ([Secondary water pump / p. 26](#))
- Check the secondary water temperature ([Water temperature / p. 22](#))

Water temperature

NOTE

The water temperature in both water circuits (primary and secondary) is measured by temperature sensors (R1, R2).

The actual temperature value can be read out from the system control table T00 ([System control table T00 / p. 15](#))

WCS internal temperature sensors

The WCS indoor unit contains 3 temperature sensors:

Tab. 9 Temperature sensors in the indoor unit

Temp.sensor	Nominal value	Function
R1	18,5C (+/-1C)	Water temperature in Secondary water circuit: Measures the temperature of the water going to gantry (output at WCS = supply side)

Temp.sensor	Nominal value	Function
R2	0C...50C	Water temperature in Primary water circuit: Measures the temperature of the water coming from the outdoor unit (input at WCS = return side).
R3	> 4C	Coolant media temp. in refrigeration circuit: Measures the temperature of the coolant media at the vaporizer output (freeze protection) for regulating the bypass valve in the refrigeration circuit.

Primary water circuit (outdoor unit)

- Check the actual water temperature in the system control table T00 ([System control table T00 / p. 15](#)) .
 - ⇒ Parameter “Primary cooling media temp.” = actual water temp. (C) in primary water circuit (return water from outdoor unit)
- Check the primary water pump ([Primary water pump / p. 25](#))
- Check the cooling fan at the outdoor unit ([Cooling fan \(outdoor unit\) / p. 26](#))
- Check capillary tube controller B1 and whether the cooling fan at the outdoor unit is switched on ([Cooling fan \(outdoor unit\) / p. 26](#))
- Check the outside air temperature next to the outdoor unit.
 - ⇒ Nominal outside air temperature: -25C....+50C

Secondary water circuit (gantry)

- Check the actual water temperature in the system control table T00 ([System control table T00 / p. 15](#)) .
 - ⇒ Parameter “WCS water temperature” = actual water temperature (C) in secondary water circuit (supply water to gantry)
 - ⇒ Nominal water temperature: 18,5 C (+/- 1C)
- Check the secondary water pump ([Secondary water pump / p. 26](#))
- Check the adjustment drive of the mixer unit (right gantry stand) ([Adjustment drive of mixer unit \(right gantry stand\) / p. 29](#))
- Check the gantry fans
- Check the refrigeration circuit ([Compressor \(refrigeration circuit\) / p. 30](#))
 - ⇒ Compressor 1 must be operating

Water pressure

NOTE

The nominal water pressure in both water circuits (primary and secondary) is 2 bar +/- 0,2 bar.

The actual water pressure at the WCS internal manometers is only displayed correctly when the WCS is switched OFF (water pumps not operating)

When the WCS (water pump) is operating the manometers will show a relative pressure difference of approx. 0,5-1 bar between the input and output side of the water pumps. The water pressure at the pump output is higher.

- Switch OFF the system.
- Open the front door of the indoor unit.

Primary water circuit

NOTE

The primary water circuit between the indoor unit (WCS cabinet) and the outdoor unit (fan cooler) contains antifreeze (40% glycol).

Therefore filling/refilling of the primary water circuit has to be executed by qualified service personnel (e.g. aircondition service).

- Check the manometers for the primary water circuit (upper section)
 - ⇒ Nominal pressure: 2 bar (+/- 0,2 bar)
 - ⇒ If pressure is too low, check for any leakage in the primary water circuit.
- Contact a qualified service personnel (e.g. aircondition service) for refilling the primary water circuit to nominal pressure.

Secondary water circuit

- Check the manometer for the secondary water circuit (lower section)
 - ⇒ Nominal pressure: 2 bar (+/- 0,2 bar)
 - ⇒ If the pressure is too low, check for any leakage in the secondary water circuit.
- Refill the secondary water circuit to nominal pressure according to the draining/filling instructions as described in the document "Replacement of Part - WCS".

Water flow

NOTE

There are no water flow sensors installed in the WCS w-a-split.

A judgement for water flow only can be made by checking the water pressure difference between pump input and output.

- Open den WCS cabinet (indoor unit)
- Check the position of the stop valve (red lever) for both water circuits
 - ⇒ The red lever must be parallel to the water hoses (= open position).
- Check the water pressure between the water pump input and output at the manometers for the according water circuit : upper manometers = primary water circuit, lower manometers = secondary water circuit
 - ⇒ Water flow is OK if the water pressure difference between water pump output and input is approx. 0,5-1 bar.
 - ⇒ If the pressure is the same on both sides there is definitely no water flow.
- Check the water pump operation ([Water pump / p. 25](#))

Water pump

NOTE

Both water pumps are normally operating permanently when the WCS is ON.

Both water pumps have an overcurrent protection switch.

Quick check

- Open the front door of the indoor unit.
- Check the water pressure between the water pump input and output at the manometers for the according water circuit : upper manometers = primary water circuit, lower manometers = secondary water circuit
 - ⇒ Water pumps are working if the water pressure at the water pump output is higher than at the pump input (pressure difference is approx. 0,5-1 bar).
 - ⇒ If the water pressure is the same on both sides the water pump is definitely not operating.
- Check the water pump status at the WCS controller board N1 ([Status indicators \(LEDs\) on N1 / p. 11](#))
 - ⇒ LED # 21 (green) must be ON if the primary water pump is active.
 - ⇒ LED # 15 (green) must be ON if primary pump overcurrent protection is inactive.
 - ⇒ LED # 19 (green) must be ON if the secondary water pump is active.
 - ⇒ LED # 14 (green) must be ON if secondary pump overcurrent protection is inactive.

Primary water pump

- Check the status of the water pump by reading out the system control table T00 ([System control table T00 / p. 15](#)).
 - ⇒ Parameter “Ventilator Y/D active” (= primary pump operation) must be marked.
 - ⇒ Parameter “not protector ventilator” (= primary pump overcurrent protection) must NOT be marked

- Check the power switch F3
 - ⇒ Switch must be in pos. "1"
- Check the contactor K3
 - ⇒ K3 must be ON
- Check the 3-phase power at F3
 - ⇒ If power is ON, K3 is active and the motor overcurrent protection is inactive
=> Replace the pump

Secondary water pump

- Check the status of the water pump by reading out the system control table T00 ([System control table T00 / p. 15](#)).
 - ⇒ Parameter "Cooling pump active" (= secondary pump operation) must be marked.
 - ⇒ Parameter "motor protector pump" (= secondary pump overcurrent protection) must NOT be marked
- Check the power switch F10
 - ⇒ Switch must be in pos. "1"
- Check the contactor K4
 - ⇒ K4 must be ON
- Check the 3-phase power at F10
 - ⇒ If power is ON, K4 is active and the motor overcurrent protection is inactive
=> Replace the pump

Cooling fan (outdoor unit)

NOTE

The power for the outdoor unit is NOT supplied by the PDC cabinet. It is supplied directly from the onsite power distribution panel into the electronic box (X4) at the indoor unit and from the electronic box (3X20) to the outdoor unit (fan cooler).

An optional "Repair switch" may be installed outside, close to the outdoor unit, if local firecode require. If the outdoor unit doesn't switch on, check for the existence an position of this switch.

If capillary tube controller B1 is installed, the fan cooler is only switched on if the primary water outlet temperature exceeds 35C.

NOTE

The air flow direction of the cooling fan at the outdoor unit is from bottom to top.

The fan motor is protected by an internal thermo contact.

- Check the bottom side of the water cooler (outdoor unit) for any items (e.g. leaves, dirt-etc.) that might handicap the air flow through the cooler.
 - ⇒ Remove any parts.
- Check the fan operation and the air flow direction.
 - ⇒ The airflow must be from bottom to top.
 - ⇒ If the airflow is reverse (top to bottom) two power phases are connected wrong.
=> Switch OFF the system and F20 (WCS electronic box) and swap 2 phases at the fan connection terminal.
 - ⇒ If the fan doesn't rotate.
=> Continue with next step.
- Check the mechanical rotation of the fan rotor by manually rotating it.
 - ⇒ If the rotor is easy rotating, continue with next step.
 - ⇒ If the rotor is blocked, check for any blocking parts.
=> Replace the fan.
- Check the 3-phase power at the connection terminal of the cooling fan with a volt meter.
 - ⇒ If 3-phase power is there, the fan is defective!
=> Replace the fan.
 - ⇒ If there is no power, continue with next step.
- Check capillary tube controller B1 (if installed) at the indoor unit with an ohmmeter. Disconnect connector 3X20 for this test. Also check the primary water outlet temperature (both tests are necessary).
 - ⇒ If the temperature is higher than 35C, the contact between clamp 1 and clamp 4 should be low impedance.
 - ⇒ If the temperature is lower than 25C, the contact between clamp 1 and clamp 4 should be high impedance.
- Check the thermo contact (motor protection) at the fan connection terminal (outdoor unit) with a volt meter.
 - ⇒ If 230V are measured at both TK pins the thermo contact (motor protection) of the fan is inactive.
=> Continue with next step
 - ⇒ If 230V are measured only at one TK pin, the motor protection of the fan is active.
=> The thermo switch will deactivate the contactor K20 and cut the fan power.
=> The thermo contact normally only gets activated if the fan is mechanically blocked (overheating) or if the fan motor has an internal problem.
=> Replace the fan
- Check the power switch F20 in the WCS electronic box (indoor unit)
 - ⇒ F20 must be ON (= position "1")
- Check the contactor K20 in the WCS electronic box (indoor unit)
 - ⇒ K20 is ON when the small compressor is activated (K1 is ON) and F20 is ON and the thermo contact of the fan is inactive (if capillary tube controller B1 is installed) if the primary water outlet temperature is above 35C.
- Check the contactor K1 (small compressor ON) in the WCS electronic box (indoor unit)
 - ⇒ K1 is ON if there is no high pressure condition in the refrigeration circuit.

Expansion reservoir

There are two expansion reservoirs installed in the indoor unit.

The smaller one is part of the secondary water circuit, the larger one is part of the primary water circuit.

NOTE

The function of the expansion reservoir only can be tested after the water circuit has been evacuated.

The manometer connector on the expansion device is at the opposite side of the water connection. It is cover with a plastic cap

Expansion reservoir (large) of primary circuit

NOTE

The primary water circuit between the indoor unit (WCS cabinet) and the outdoor unit (fan cooler) contains antifreeze (40% glycol).

Therefore the replacement of parts in the primary water circuit that require a draining and refilling of the water circuit has to be executed by qualified service personnel (e.g. air-condition service).

Antifreeze-enhanced cooling water must be disposed accordingly.

- Switch OFF the system.
- Open the front door at the indoor unit.
- Remove the right side cover of the WCS cabinet.
- Drain and evacuate the relevant water circuit according to the draining/filling instructions as described in the document "Replacement of parts - WCS".
- Remove the plastic cap from the expansion device and connect the manometer to the connector.
 - ⇒ Nominal pressure (large expansion device): 1,5 bar (= primary water circuit)
 - ⇒ Nominal pressure (small expansion device): 0,5 bar (= secondary water circuit)
- If the pressure is 0 bar, the expansion device is defective.
 - ⇒ Replace the expansion device

Expansion reservoir (small) of secondary circuit

- Switch OFF the system.
- Open the front door at the indoor unit.
- Drain and evacuate the relevant water circuit according to the draining/filling instructions as described in the document "Replacement of parts - WCS".

NOTE

The primary water circuit between the indoor unit (WCS cabinet) and the outdoor unit (fan cooler) contains antifreeze (40% glycol).

Therefore the replacement of parts in the primary water circuit that require a draining and refilling of the water circuit has to be executed by qualified service personnel (e.g. air-condition service).

Antifreeze-enhanced cooling water must be disposed accordingly.

- Remove the plastic cap from the expansion device and connect the manometer to the connector.
 - ⇒ Nominal pressure (large expansion device): 1,5 bar (= primary water circuit)
 - ⇒ Nominal pressure (small expansion device): 0,5 bar (= secondary water circuit)
- If the pressure is 0 bar, the expansion device is defective.
 - ⇒ Replace the expansion device

Adjustment drive of mixer unit (right gantry stand)

The adjustment drive of the mixer unit in the right gantry stand is part of the secondary water circuit. It regulates the supply of chilled water into the gantry depending on the required cooling power (heat dissipation) of the gantry.

NOTE

When the system is switched ON the adjustment drive of the mixer is moved to Position 0 to fully bypass the gantry water circuit.

After approx. 3 minutes the WCS starts operation and the adjustment drive regulates the gantry water circuit. Therefore the adjustment drive must be moving away from Pos.0.

This is an easy test to see if the adjustment drive works.

- Switch OFF the system.
- Remove the cover at the right gantry stand.
- Turn the button at the adjustment drive by hand to Position 5
- Switch ON the system.
- Check the Position of the adjustment drive button
 - ⇒ The adjustment drive must be in Pos. 0
- Wait for approx. 3 minutes after switching ON the system.
 - ⇒ The adjustment drive must have moved away from Pos. 0.
- If the adjustment drive has not moved at all, check the cable connection to the adjustment drive.
 - ⇒ If cable connection is OK, replace the adjustment drive.

Compressor (refrigeration circuit)

NOTE

The refrigeration/chiller circuit (indoor unit) contains a coolant media.

Only qualified service personnel (e.g. aircondition service) is allowed to perform service and repair work in the refrigeration circuit.

=> Contact a qualified local aircondition service for service work.

=> A recommended service contact list can be found in the spare parts catalogue under the WCS component.

- Check the compressor status at the WCS controller board N1 ([Status indicators \(LEDs\) on N1 / p. 11](#))
 - ⇒ LED # 17 (green) must be ON if the compressor 1 (small) is active. (Compressor 1 is normally always operating)
 - ⇒ LED # 18 (green) must be ON if the compressor 2 (large) is active. (Compressor 2 is only operating if the required cooling power exceeds 3,5 kW)
- Check the contactor K1 in the WCS electronic-box.
 - ⇒ K1 must be ON
- Check the status of the refrigeration circuit by reading out the system control table T00 ([System control table T00 / p. 15](#)).
 - ⇒ Parameter “cooling system defective” (= refrigeration circuit error) should be marked.
 - ⇒ Contact a qualified aircondition service for further troubleshooting the refrigeration circuit

High pressure condition (refrigeration circuit)

NOTE

A high pressure condition can be caused if the return water temperature from the outdoor unit is too high. This condition can be caused either by too hot outside air, defective cooling fan or defective primary water pump.

- Check the status of LED # 11 at the WCS control board N1 ([Status indicators \(LEDs\) on N1 / p. 11](#))
 - ⇒ LED # 11 (green) must be ON if there is no high pressure condition. If the LED # 11 is OFF, continue with the next step.
- Check the outside air temperature at the outdoor unit
 - ⇒ Max. outside air temperature: + 50°C If the air temperature is OK, continue with the next step.
- Check the fan operation at the outdoor unit ([Cooling fan \(outdoor unit\) / p. 26](#))
 - ⇒ If the fan is operating, continue with the next step

- Check the primary water pump operation ([Primary water pump / p. 25](#))
 - ⇒ If the pump is operating, there is a problem in the refrigeration circuit.
 - ⇒ Contact a qualified aircondition service for further troubleshooting the refrigeration circuit

Low pressure condition (refrigeration circuit)

NOTE

A low pressure condition is most probably caused by a leakage in the refrigeration circuit of the WCS system.

=> Loss of coolant media!

NOTE

The refrigeration/chiller circuit (indoor unit) contains a coolant media.

Only qualified service personnel (e.g. aircondition service) is allowed to perform service and repair work in the refrigeration circuit.

=> Contact a qualified local aircondition service for service work.

=> A recommended service contact list can be found in the spare parts catalogue under the WCS component.

- Check the LED # 10 (low pressure condition in the refrigeration circuit) at the WCS controller board N1 ([Status indicators \(LEDs\) on N1 / p. 11](#))
 - ⇒ LED # 10 (green) must be ON if there is no low pressure condition. If LED 10 is off, there is definitely a problem in the refrigeration circuit
- Check the status of the refrigeration circuit by reading out the system control table T00 ([System control table T00 / p. 15](#)).
 - ⇒ Parameter “cooling system defective” (= refrigeration circuit error) must be marked.
 - ⇒ Contact a qualified aircondition service for troubleshooting the refrigeration circuit

WCS controller board N1

Quick check

- Pull the cable from connector X3 (MCU connection) at the N1 board
- Push the switch S1 at the N1 board
 - ⇒ If the red LED # 16 comes ON and goes OFF after some seconds the N1 board should be OK.
- Connect cable to connector X3 again.

NOTE

Do not push switch S1 during WCS operation!

This will cause an undefined status of the WCS and will stop the WCS operation.

A restart of the WCS only will be possible after a delaytime of 3 minutes.

Chapter	Change	Reason
2	Added “repair switch”	515289

